CCIT Report No. 382

Noise Cancellation with Static Mixtures of a Nonstationary Signal and Stationary Noise

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ABSTRACT

We address the problem of canceling a stationary noise component from its static mixtures with a nonstationary signal of interest. This problem can be treated from two different perspectives, using different processing tools, both based on second-order statistics. The first approach is the Blind Source Separation (BSS) approach, which is aimed at estimating the mixing parameters via approximate joint diagonalization of estimated correlation matrices. Proper exploitation of the nonstationary nature of the desired signal in contrast to the stationarity of the noise signal, allows special parameterization of the joint diagonalization problem in terms of a nonlinear weighted least squares (WLS) problem. The second approach is a denoising approach, which translates into direct estimation of just one of the mixing coefficients via solution of a linear WLS problem, followed by the use of this coefficient to create a noise-only signal to be properly eliminated from the mixture. Under certain assumptions, the BSS approach is asymptotically optimal, yet computationally more intense than the suboptimal denoising approach, since it involves an iterative solution of a nonlinear WLS, whereas the latter only requires a closed-form linear LS solution. We analyze and compare the performance of the two approaches, and provide some simulation results which confirm our analysis.